

SECOND EDITION

STEVEN B. ACHELIS

Technical Analysis from



A to Z

COVERS EVERY TRADING
TOOL . . . FROM THE
ABSOLUTE BREADTH INDEX
TO THE ZIG ZAG



Technical Analysis from A to Z

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**Covers Every Trading Tool . . .
from the Absolute Breadth Index to the Zig Zag**

Second Edition

Steven B. Achelis

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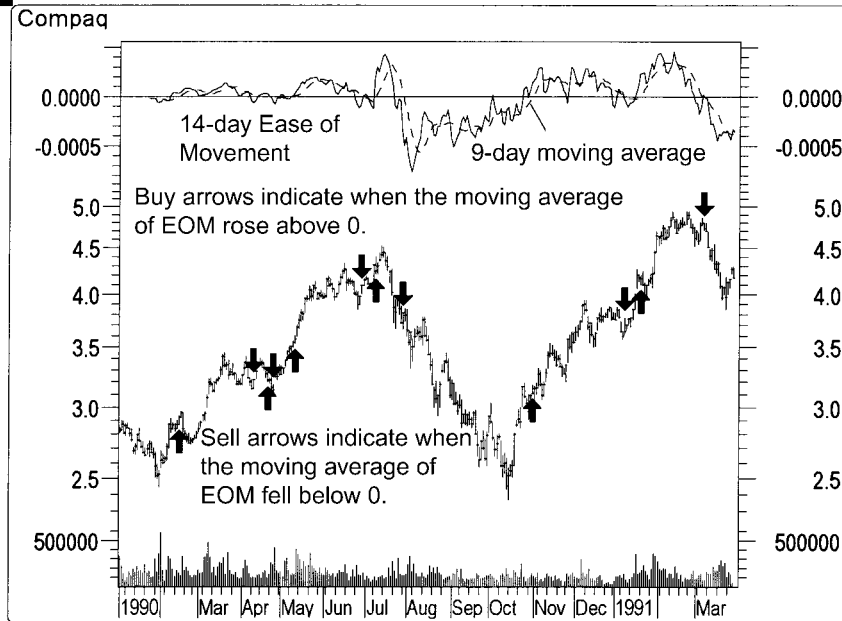
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Example

Figure 79 shows Compaq and a 14-day Ease of Movement indicator. A 9-day moving average was plotted on the Ease of Movement indicator. "Buy" and "sell" arrows indicate when the moving average crossed zero.

FIGURE 79



Calculation

In his book, Arms describes two methods of calculating the Ease of Movement indicator: the "exact method" as presented here and the "simplified method" where the "high + low" is expressed in eighths.

The first step in calculating the Ease of Movement indicator is to determine the Midpoint Move as shown below.

$$MM = \left(\frac{High + Low}{2} \right) - \left(\frac{Previous High + Previous Low}{2} \right)$$

Next, calculate the high-low Box Ratio.

$$Box Ratio = \frac{Volume \text{ (in 10,000s)}}{High - Low}$$

TABLE 23

A	B	C	D	EASE OF MOVEMENT		G	H	I	J
				E	F				
Date	High	Low	Volume	High plus Low	Column E divided by 2	Column F minus Previous Column F	High minus Low	Col D / 10,000 divided by Column H	Column G divided by Column I
07/01/99	23.7500	23.0000	125,733	46.7500	23.3750	N/A	0.7500	16.7644	N/A
07/02/99	23.7500	23.1875	83,819	46.9375	23.4688	0.0938	0.5625	14.9012	0.0063
07/06/99	23.7500	23.2500	111,390	47.0000	23.5000	0.0313	0.5000	22.2780	0.0014
07/07/99	25.0625	23.5000	211,366	48.5625	24.2813	0.7813	1.5625	13.5274	0.0578
07/08/99	26.2500	25.0000	240,664	51.2500	25.6250	1.3438	1.2500	19.2531	0.0698
07/09/99	26.8750	25.8125	219,933	52.6875	26.3438	0.7188	1.0625	20.6996	0.0347
07/12/99	27.0000	25.8750	155,943	52.8750	26.4375	0.0938	1.1250	13.8616	0.0068
07/13/99	26.8750	25.7500	138,913	52.6250	26.3125	-0.1250	1.1250	12.3478	-0.0101
07/14/99	28.0000	26.3125	226,220	54.3125	27.1563	0.8438	1.6875	13.4056	0.0629
07/15/99	28.1875	27.3750	164,528	55.5625	27.7813	0.6250	0.8125	20.2496	0.0309
07/16/99	27.6875	27.1250	132,053	54.8125	27.4063	-0.3750	0.5625	23.4761	-0.0160
07/19/99	27.1875	26.0000	109,900	53.1875	26.5938	-0.8125	1.1875	9.2547	-0.0878
07/20/99	26.2500	25.1875	138,313	51.4375	25.7188	-0.8750	1.0625	13.0177	-0.0672
07/21/99	26.5000	25.3750	143,421	51.8750	25.9375	0.2188	1.1250	12.7485	0.0172
07/22/99	26.0000	24.8750	106,053	50.8750	25.4375	-0.5000	1.1250	9.4269	-0.0530
07/23/99	25.8750	24.3125	141,425	50.1875	25.0938	-0.3438	1.5625	9.0512	-0.0380
07/26/99	25.3750	24.2500	96,921	49.6250	24.8125	-0.2813	1.1250	8.6152	-0.0326
07/27/99	25.5000	24.7500	93,208	50.2500	25.1250	0.3125	0.7500	12.4277	0.0251

The Ease of Movement (EMV) indicator is then calculated from the Midpoint Move and Box Ratio.

$$EMV = \frac{\text{Midpoint Move}}{\text{Box Ratio}}$$

Table 23 illustrates the calculation of the Ease of Movement indicator.

- Column E is the sum of the high plus the low.
- Column F is the value in Column E divided by 2. This is the midpoint.
- Column G is calculated by subtracting Column F from the previous day's Column F.
- Column H is the value of the high minus the low.
- Column I is calculated by first dividing the volume by 10,000. Next, divide this result by Column H. This is the Box Ratio.
- Column J is Column G divided by Column I. This is the Ease of Movement.
- The raw Ease of Movement value is usually smoothed with a moving average.

E

EFFICIENT MARKET THEORY

Overview

The Efficient Market Theory says that security prices correctly and almost immediately reflect all information and expectations. It says that you cannot consistently outperform the stock market due to the random nature in which information arrives and the fact that prices react and adjust almost immediately to reflect the latest information. Therefore, it assumes that at any given time, the market correctly prices all securities. The result, or so the theory advocates, is that securities cannot be overpriced or underpriced for a long enough period of time to profit therefrom.

The theory holds that since prices reflect all available information, and since information arrives in a random fashion, there is little to be gained by any type of analysis, whether fundamental or technical. It assumes that every piece of information has been collected and processed by thousands of investors and this information (both old and new) is correctly reflected in the price. Returns cannot be increased by studying historical data, either fundamental or technical, since past data will have no effect on future prices.

The problem with both of these concepts is that *many* investors base their expectations on past prices (whether using technical indicators, fundamental track records, overbought/oversold conditions, industry trends, etc). Since investors' expectations control prices, it seems obvious that past prices do have a significant influence on future prices.

E

ELLIOTT WAVE THEORY

Overview

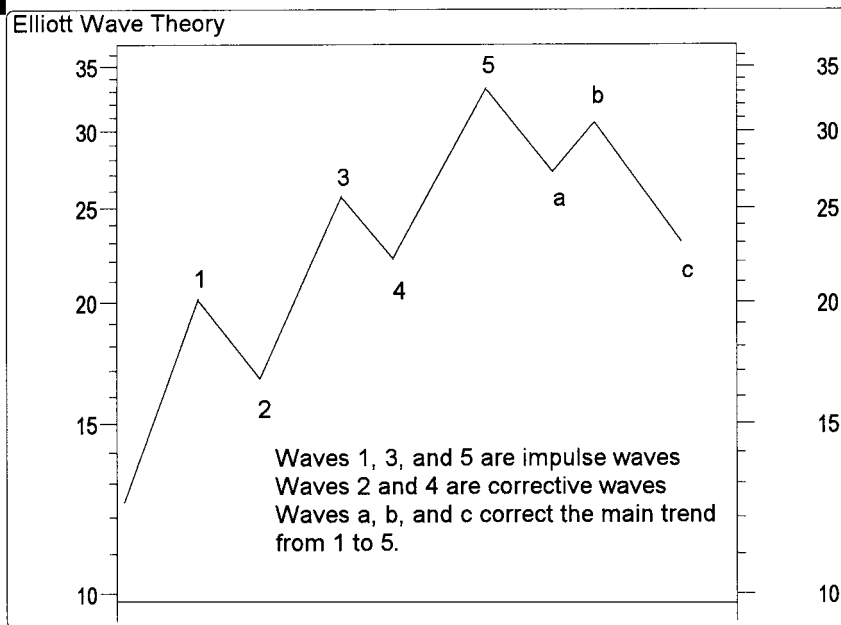
The Elliott Wave Theory is named after Ralph Nelson Elliott. Inspired by the Dow Theory (see page 123) and by observations found throughout nature, Elliott concluded that the movement of the stock market could be predicted by observing and identifying a repetitive pattern of waves. In fact, Elliott believed that all human activities, not just the stock market, were influenced by these identifiable series of waves.

With the help of C. J. Collins, Elliott's ideas received the attention of Wall Street in a series of articles published in 1939 in *Financial World* magazine. During the 1950s and 1960s (after Elliott's passing), his work was advanced by Hamilton Bolton. In 1960, Bolton wrote *Elliott Wave Principle: A Critical Appraisal*. This was the first significant work since Elliott's passing. In 1978, Robert Prechter and A. J. Frost collaborated to write *Elliott Wave Principle*.

Interpretation

The underlying forces behind the Elliott Wave Theory are those of building up and tearing down. The basic concepts of the Elliott Wave Theory are listed below.

1. Action is followed by reaction.
2. There are five waves in the direction of the main trend followed by three corrective waves (a "5-3" move).
3. A 5-3 move completes a cycle. This 5-3 move then becomes 2 subdivisions of the next higher 5-3 wave.
4. The underlying 5-3 pattern remains constant, though the time span of each may vary.

FIGURE 80

The basic pattern is made up of 8 waves (5 up and 3 down) that are labeled 1, 2, 3, 4, 5, a, b, and c in Figure 80.

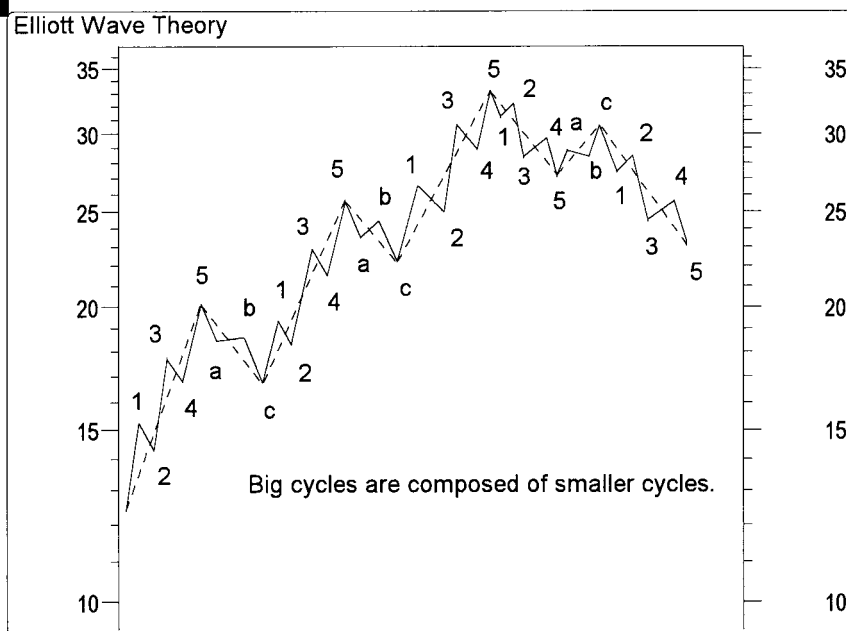
Waves 1, 3, and 5 are called "impulse waves." Waves 2 and 4 are called "corrective waves." Waves a, b, and c correct the main trend made by waves 1 through 5.

The main trend is established by waves 1 through 5 and can be either up or down. Waves a, b, and c always move in the opposite direction of waves 1 through 5.

Elliott Wave Theory holds that each wave within a wave count contains a complete 5-3 wave count of a smaller cycle. The longest wave count is called the "Grand Supercycle." Grand Supercycle waves are comprised of Supercycles, and Supercycles are comprised of Cycles. This process continues into Primary, Intermediate, Minute, Minuette, and Sub-minute waves.

Figure 81 shows how 5-3 waves are comprised of smaller cycles. This chart contains the identical pattern shown in the preceding chart (now displayed using dotted lines), but the smaller cycles are also displayed. For example, you can see that the impulse wave labeled "1" in the preceding chart is comprised of 5 smaller waves.

Fibonacci numbers (see page 141) provide the mathematical foundation for the Elliott Wave Theory. Briefly, the Fibonacci number sequence is made

FIGURE 81

simply by starting at 1 and adding the previous number to arrive at the new number (i.e., $0 + 1 = 1$, $1 + 1 = 2$, $2 + 1 = 3$, $3 + 2 = 5$, $5 + 3 = 8$, $8 + 5 = 13$, etc.). Each of the cycles that Elliott defined is comprised of a total wave count that falls within the Fibonacci number sequence. For example, Figure 81 shows 2 primary waves (an impulse wave and a corrective wave), 8 intermediate waves (the 5-3 sequence shown in the first chart), and 34 minute waves (as labeled). The numbers 2, 8, and 34 fall within the Fibonacci numbering sequence.

Elliott Wave practitioners use their determination of the wave count in combination with the Fibonacci numbers to predict the time span and magnitude of future market moves ranging from minutes and hours to years and decades.

Elliott Wave practitioners generally agree that the most recent Grand Supercycle began in 1932 and that the final fifth wave of this cycle began at the market bottom in 1982. However, much disparity has manifested since 1982. Many heralded the arrival of the October 1987 crash as the end of the cycle. The strong recovery that has since followed has caused them to reevaluate their wave counts. Herein lies the weakness of the Elliott Wave Theory—its predictive value is dependent on an accurate wave count. Determining where one wave starts and another wave ends can be extremely subjective.

ENVELOPES (TRADING BANDS)

Overview

An envelope is comprised of two moving averages (see page 203). One moving average is shifted upward and the second is shifted downward.

Interpretation

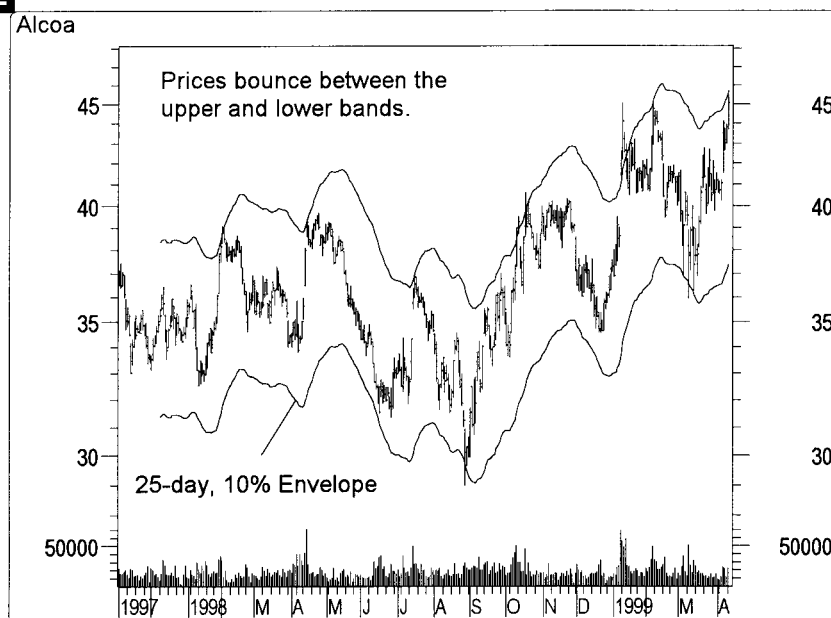
Envelopes define the upper and lower boundaries of a security's normal trading range. A sell signal is generated when the security reaches the upper band, whereas a buy signal is generated at the lower band. The optimum percentage shift depends on the volatility of the security—the more volatile, the larger the percentage.

The logic behind envelopes is that overzealous buyers and sellers push the price to the extremes (i.e., the upper and lower bands), at which point the prices often stabilize by moving to more realistic levels. This is similar to the interpretation of Bollinger Bands (see page 71).

Example

Figure 82 displays Alcoa with a 10-percent envelope of a 25-day exponential moving average. You can see how Alcoa's price tended to bounce off the bands rather than penetrate them.

FIGURE 82



Calculation

Envelopes are calculated by plotting two moving averages that have been shifted up and down. For example, the upper band might be shifted up 3 percent and the lower band shifted down 3 percent.

Table 24 illustrates the calculation of Envelopes. In this example the Envelopes are calculated using a 5-week simple moving average with a 3-percent shift.

- Column C is a 5-period simple moving average of the close. It is calculated by adding the closing price for the preceding 5 periods and dividing by 5.
- Column D is the value of Column C multiplied by 1.03 (i.e., 100 percent plus a 3-percent shift). This is the upper band.
- Column E is the value of Column C multiplied by 0.97 (i.e., 100 percent minus a 3-percent shift). This is the lower band.

TABLE 24

		ENVELOPES		
A	B	C	D	E
Date	Close	5-week simple moving average	Column C multiplied by 1.03	Column C multiplied by 0.97
11/15/91	18.594			
11/22/91	18.656			
11/29/91	18.563			
12/06/91	19.563			
12/13/91	19.188	18.913	19.480	18.345
12/20/91	19.969	19.188	19.763	18.612
12/27/91	21.406	19.738	20.330	19.145
01/03/92	20.781	20.181	20.787	19.576
01/10/92	20.813	20.431	21.044	19.818

EQUIVOLUME

Overview

Equivolume displays prices in a way that emphasizes the relationship between price and volume. Equivolume was developed by Richard W. Arms, Jr. and is further explained in his book *Volume Cycles in the Stock Market*.

Table 55 illustrates the calculation of the Performance indicator.

- Column C is the close (Column B) minus the first closing price. For example, on 01/13/99 Column C is the close (26.7500) minus the first close (26.5625).
- Column D is the value in Column C divided by the *first* value in Column B and multiplied by 100. This is the Performance indicator.

TABLE 55

PERFORMANCE			
A	B	C	D
Date	Close	Col B minus first value in Col B	Col C divided by First value in Col B times 100
01/04/99	26.5625		
01/05/99	25.8125	-0.7500	-2.82%
01/06/99	25.8750	-0.6875	-2.59%
01/07/99	27.5000	0.9375	3.53%
01/08/99	28.4062	1.8437	6.94%
01/11/99	27.5312	0.9687	3.65%
01/12/99	27.4688	0.9063	3.41%
01/13/99	26.7500	0.1875	0.71%
01/14/99	25.6875	-0.8750	-3.29%
01/15/99	28.1562	1.5937	6.00%

P

POINT AND FIGURE

Overview

Point and Figure (P&F) charts differ from traditional price charts in that they completely disregard the passage of time and display only changes in prices. Rather than having price on the y-axis and time on the x-axis, P&F charts display price changes on both axes. This is similar to Kagi (see page 164), Renko (see page 302), and Three Line Break (see page 330) charts.

Interpretation

P&F charts display the underlying supply and demand of prices. A column of X's shows that demand is exceeding supply (a rally); a column of O's shows that supply is exceeding demand (a decline); and a series of short columns shows that supply and demand are relatively equal.

Several chart patterns (see page 245) regularly appear in P&F charts. These include Double Tops and Bottoms, Bullish and Bearish Signal Formations, Bullish and Bearish Symmetrical Triangles, Triple Tops and Bottoms, etc. It is beyond the scope of this book to fully explain all of these patterns.

Example

P&F charts focus only on price action. Looking at the P&F chart in Figure 140, you can see that prices were initially contained between a support level at 56.50 and a resistance level at 60.50. When prices broke above the resistance level at 60.50 (the long column of X's), that level became the new support level. This new support level eventually failed (the long column of O's), prices retested the support at 56.50, made a small rally, and then fell below the 56.50 support level.

FIGURE 140

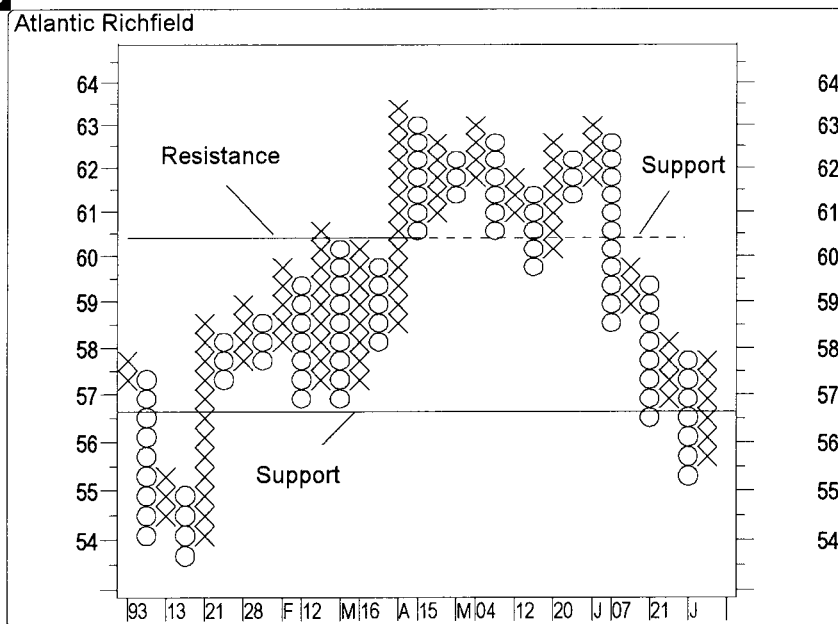
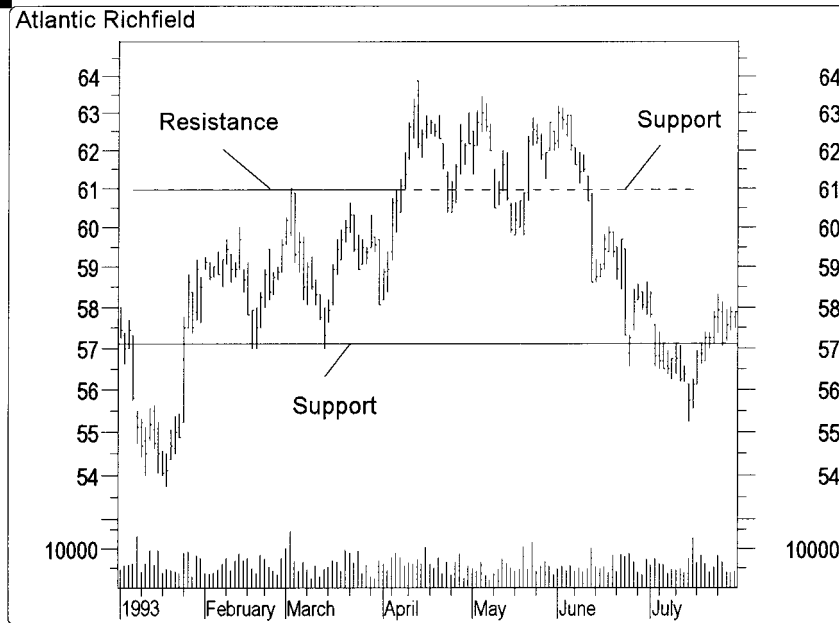


Figure 141 shows the same pricing information as the preceding P&F chart. You can see that the support and resistance levels are also identifiable in this bar chart, but the P&F chart made it much easier to identify them.

FIGURE 141

Calculation

P&F charts display an "X" when prices rise by the "box size" (a value you specify) and display an "O" when prices fall by the box size. Note that no X's or O's are drawn if prices rise or fall by an amount that is less than the box size.

Each column can contain either X's or O's, but never both. In order to change columns (e.g., from an X column to an O column), prices must reverse by the "reversal amount" (another value you specify) multiplied by the box size. For example, if the box size is 3 points and the reversal amount is 2 boxes, then prices must reverse direction 6 points (3 multiplied by 2) in order to change columns. If you are in a column of X's, the price must fall 6 points to change to a column of O's. If you are in a column of O's, the price must rise 6 points to change to a column of X's.

The changing of columns identifies a change in the trend of prices. When a new column of X's appears, it shows that prices are rallying higher. When a new column of O's appears, it shows that prices are moving lower.

Because prices must reverse direction by the reversal amount, the minimum number of X's or O's that can appear in a column is equal to the "reversal amount."

The common practice is to use the high and low prices (not just the close) to decide if prices have changed enough to display a new box.

POLARIZED FRACTAL EFFICIENCY

Overview

The Polarized Fractal Efficiency (PFE) indicator, developed by Hans Hannula, applies the laws of fractal geometry and chaos theory to the markets. PFE uses mathematics to determine how efficiently a price travels between two points. The more linear and efficient the price movement, the shorter the distance the prices must travel between two points.

Interpretation

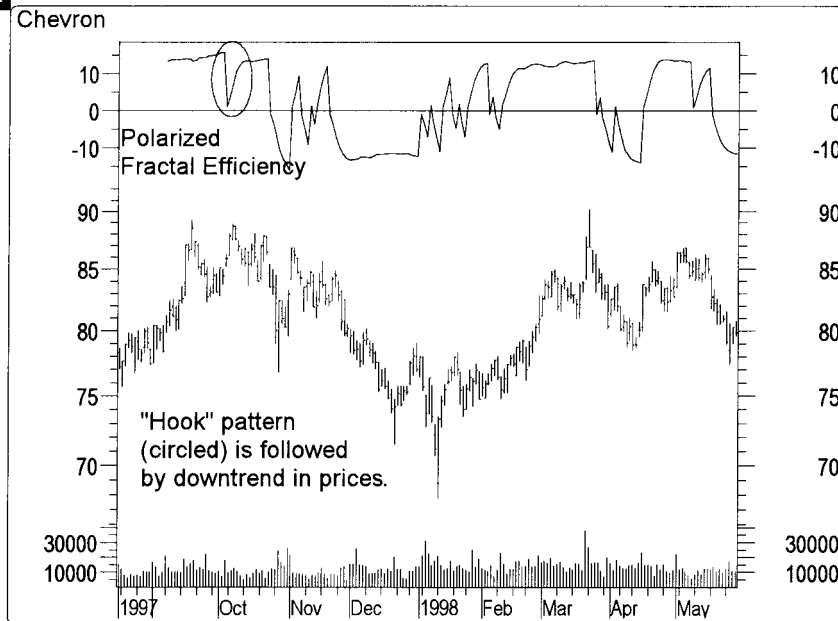
The primary use of the PFE indicator is as a measure of how trendy or congested price action is. PFE readings above zero mean that the trend is up. The higher the reading, the “trendier” and more efficient the upward movement. PFE readings below zero mean that the trend is down. The lower the reading, the “trendier” and more efficient the downward movement. Readings around zero indicate choppy, less efficient movement, with a balance between the forces of supply and demand.

Several interesting phenomenon have been observed by Mr. Hannula:

- Indexes (particularly the OEX) tend to have a maximum PFE (both plus and minus) of about 43 percent.
- The middle region (around zero) is a balance between supply and demand and therefore a congestion point.
- A hooking pattern often occurs right before the end of an efficient period. This pattern occurs when the PFE appears to have reached its maximum, it turns in the opposite direction towards zero, and then makes one last attempt at maximum efficiency. Trades can be entered in the opposite direction, with a stop just beyond the extreme of the hook. Stay with the trade all the way to the other extreme, unless it slows around the zero line. If it slows around zero, exit the trade and wait for a new maximum efficiency entry.

Example

Figure 142 of Chevron shows a 14-day PFE with three smoothing periods. This chart shows the “hooking” pattern described previously. Prices were trending up during an efficient period, then the PFE made a run towards zero, then made a last attempt at maximum efficiency, then turned down for good, marking a turning point in Chevron’s price.

FIGURE 142

Calculation

The calculations for the Polarized Fractal Efficiency are quite complex and beyond the scope of this book. For further information, refer to the January 1994 issue of *Technical Analysis of Stocks and Commodities*.

P

POSITIVE VOLUME INDEX

Overview

The Positive Volume Index (PVI) focuses on days when the volume increased from the previous day. The premise is that the "crowd" takes positions on days when volume increases.

Interpretation

Interpretation of the PVI assumes that on days when volume increases, the crowd-following, "uninformed" investors are in the market. Conversely, on

days with decreased volume, the “smart money” is quietly taking positions. Thus, the PVI displays what the not-so-smart money is doing. (The Negative Volume Index, page 214, displays what the smart money is doing.) Note, however, that the PVI is not a contrarian indicator. Even though the PVI is supposed to show what the not-so-smart money is doing, it still trends in the same direction as prices.

Because rising prices are usually associated with rising volume, the PVI usually trends upward.

Table 56 summarizes NVI and PVI data from 1941 through 1975 as explained in *Stock Market Logic* by Norman Fosback.

TABLE 56

Indicator	Indicator relative to 1-year moving average	Probability that bull market is in progress	Probability that bear market is in progress
NVI	Above	96%	4%
PVI	Above	79%	21%
NVI	Below	47%	53%
PVI	Below	33%	67%

As you can see, the NVI is excellent at identifying bull markets (i.e., when the NVI is above its 1-year moving average) and the PVI is pretty good at identifying bull markets (when the PVI is above its moving average) and bear markets (i.e., when the PVI is below its moving average).

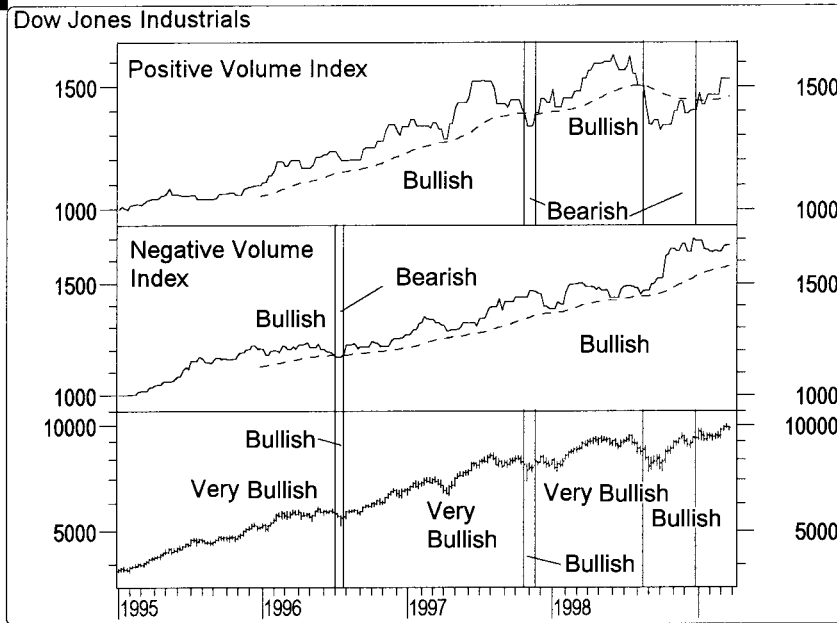
Example

Figure 143 shows the NVI, the PVI, and the Dow Jones Industrial Average over a 4-year period (weekly data). The NVI and PVI indicators are labeled bullish or bearish, depending on if they were above or below their 52-week moving averages.

The Dow Jones Industrial Average is labeled as bullish when either the NVI or PVI was above its moving average, and as very bullish when both the indicators were both above their moving averages.

Calculation

The calculation of the PVI is based on whether the current period’s volume is greater than, or less than, the previous period’s volume.

FIGURE 143

If the current volume is greater than the previous period's volume, then:

$$PVI = \text{Previous PVI} + \left(\frac{\text{Close} - \text{Previous Close}}{\text{Previous Close}} * \text{Previous PVI} \right)$$

However, if the current volume is less than or equal to the previous period's volume, then:

$$PVI = \text{Previous PVI}$$

Table 57 illustrates the calculation of the PVI.

- On the first day, Column G is "seeded" with 1,000. This gives us an initial PVI. (The value of the "seed" isn't important, as long as it isn't zero.)
- Column D is the change in price from the previous period. It is calculated by subtracting the previous close (Column B) from the current close.
- Column E is the percent change in price. It is calculated by dividing Column D (the change in price) by the previous close (the previous row in Column B).
- Column F is the value in Column E (i.e., the 1-period percent change) multiplied by the previous value in Column G (i.e., the previous PVI).

TABLE 57

POSITIVE VOLUME INDEX						
A	B	C	D	E	F	G
Date	Close	Volume	Close minus Previous Close	Column D divided by Previous Close	Column E multiplied by Previous Col G	PVI (see text)
01/04/88	6.5313	10,500				1,000.0000
01/05/88	6.5625	6,492	0.0312	0.0048	4.7770	1,000.0000
01/06/88	6.4688	6,540	-0.0937	-0.0143	-14.2781	985.7219
01/07/88	6.4375	8,924	-0.0313	-0.0048	-4.7695	980.9524
01/08/88	6.2188	5,416	-0.2187	-0.0340	-33.3257	980.9524
01/11/88	6.2500	4,588	0.0312	0.0050	4.9215	980.9524
01/12/88	6.1250	16,236	-0.1250	-0.0200	-19.6190	961.3333
01/13/88	6.1563	2,864	0.0313	0.0051	4.9126	961.3333
01/14/88	6.2500	3,080	0.0937	0.0152	14.6317	975.9650
01/15/88	6.3125	5,312	0.0625	0.0100	9.7597	985.7247
01/18/88	6.4688	12,456	0.1563	0.0248	24.4069	1,010.1316
01/19/88	6.2813	3,020	-0.1875	-0.0290	-29.2790	1,010.1316
01/20/88	6.1250	12,228	-0.1563	-0.0249	-25.1355	984.9961
01/21/88	6.1875	5,248	0.0625	0.0102	10.0510	984.9961
01/22/88	6.1563	6,632	-0.0312	-0.0050	-4.9668	980.0293
01/25/88	6.2813	4,104	0.1250	0.0203	19.8989	980.0293
01/26/88	6.2188	4,464	-0.0625	-0.0100	-9.7515	970.2779
01/27/88	6.1563	6,212	-0.0625	-0.0101	-9.7515	960.5264
01/28/88	6.1875	3,836	0.0312	0.0051	4.8679	960.5264
01/29/88	6.3750	7,640	0.1875	0.0303	29.1069	989.6333
02/01/88	6.2813	3,544	-0.0937	-0.0147	-14.5457	989.6333
02/02/88	6.2813	4,660	0.0000	0.0000	0.0000	989.6333

Calculation

The formula for Standard Deviation is as follows:

$$\sqrt{\frac{\sum_{j=1}^n (Close_j - n\text{-period SMA of Close})^2}{n}}$$

Where:

- SMA = Simple Moving Average
- n = Number of time periods

Standard Deviation is derived by calculating an n -period simple moving average of the data item (i.e., the closing price or an indicator), summing the squares of the difference between the data item and its moving average over each of the preceding n time periods, dividing this sum by n , and then calculating the square root of this result.

The Standard Deviation can be calculated using the *stddev()* spreadsheet function (which calculates the Standard Deviation) or manually as shown below in Table 72, which illustrates the calculation of the 5-period Standard Deviation of closing prices.

- Column C is a 5-day simple moving average of closing prices. It is calculated by adding the preceding 5 days of closing prices and dividing by 5.
- Column D subtracts each of the last 5 closes from today's simple moving average (e.g., the moving average minus the close 5 days ago, the moving average minus the close 4 days ago, etc.). Each of these 5 numbers is squared, and then the five squared values are summed.
- Column E divides Column D by the number of periods (e.g., 5). It then takes the square root of this result. This is the Standard Deviation.

TABLE 72

STANDARD DEVIATION				
A	B	C	D	E
Date	Close	5-day SMA of Column B	Sum of squares of Col B – Col C last 5 periods (see text)	Square root of Column D divided by 5
01/04/99	21.4375			
01/05/99	21.6875			
01/06/99	22.1250			
01/07/99	21.5625			
01/08/99	21.8125	21.7250	0.2781	0.2358
01/11/99	21.4375	21.7250	0.2781	0.2358
01/12/99	21.5000	21.6875	0.3203	0.2531
01/13/99	21.1875	21.5000	0.2031	0.2016
01/14/99	20.6875	21.3250	0.7063	0.3758
01/15/99	20.9375	21.1500	0.4656	0.3052

STANDARD DEVIATION CHANNEL

Overview

Standard Deviation Channels are calculated by plotting two parallel lines above and below an n -period Linear Regression Trendline (see page 176). The lines are plotted a specified number of standard deviations away from the Linear Regression Trendline.

Interpretation

Statistical analysis states that approximately 67 percent of future price movement should be contained within one standard deviation and approximately 95 percent within two standard deviations. However, this assumes random, trendless data. Since most markets show overwhelming evidence of non-random, trending behavior, the 67-percent and 95-percent values tend to be low (i.e., prices are even more likely to be contained). In addition, Standard Deviation Channels incorporate the current trend (as shown by the middle Linear Regression Trendline), which makes prices even more likely to be contained by the channels.

As with other channels, price movement should be expected to return to the Linear Regression Trendline. Movement outside the upper or lower channel on heavy volume may indicate a change in trend.

Standard Deviation Channels can also be used in conjunction with other indicators. An effective combination would be a buy signal from another indicator combined with a price near the lower channel (and vice versa).

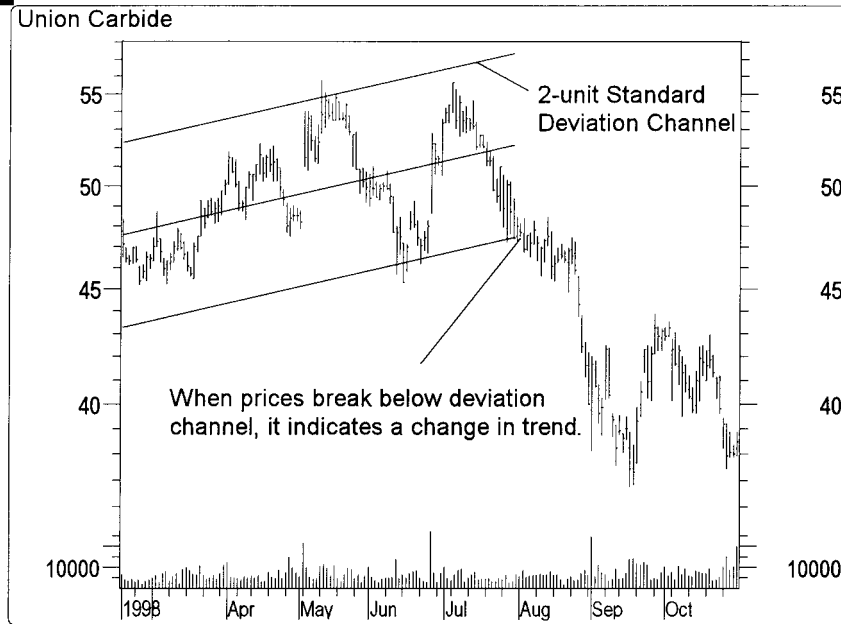
Example

S

The Standard Deviation Channels of Union Carbide in Figure 169 are placed two deviations away from the Linear Regression Trendline. We would expect 95 percent of future price movement to be contained within the bands. When the price breaks convincingly below the bottom band, a change in trend is indicated.

Calculation

The center line of the Standard Deviation Channel is a Linear Regression Trendline (see page 176). The upper channel is plotted the specified number of standard deviations (see page 308) above the center line. The lower channel is plotted the specified number of standard deviations below the center line.

FIGURE 169

STANDARD ERROR

Overview

Standard Error measures how closely prices congregate around a Linear Regression Trendline (see page 176). It quantifies how closely the actual prices are to the prices estimated by a Linear Regression Trendline. The closer prices are to the Linear Regression Trendline and the higher the r-Squared value (see page 282), the stronger the trend.

For example, if each day's closing price was equal to that day's Linear Regression Trendline value, then the Standard Error would be zero. The more variance around the Linear Regression Trendline, the larger the Standard Error and the less reliable the trend.

Interpretation

High Standard Error values indicate that the security's prices are very volatile around the Linear Regression Trendline. Changes in the prevailing trend are usually preceded by a rapidly increasing Standard Error.

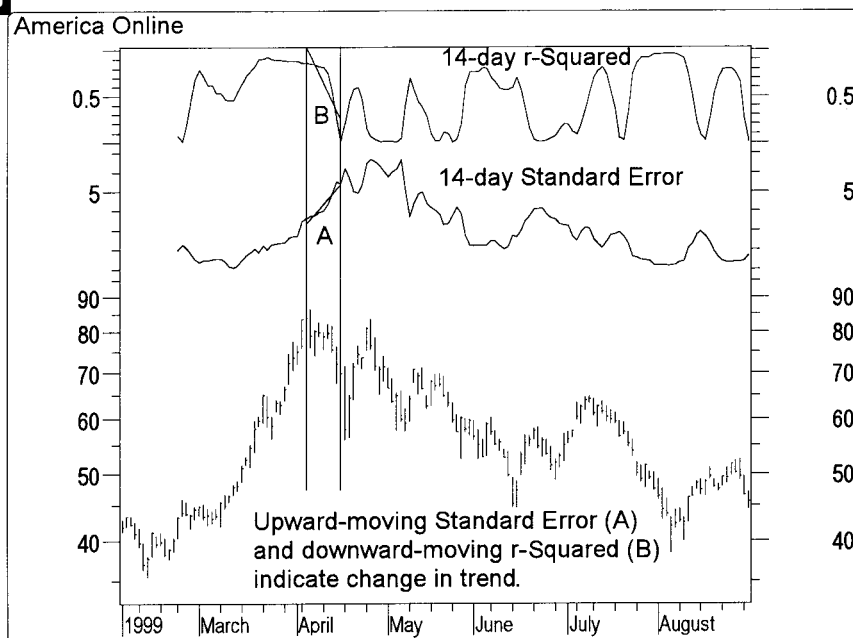
Standard Error can be used effectively in combination with the r-Squared indicator. Changes in trend are often signaled by a high downward-moving r-Squared combined with a low upward-moving Standard Error indicating a weakening trend and increasing volatility around the trend, or a low upward-moving r-Squared and a high downward-moving Standard Error indicating a strengthening trend and decreasing volatility around the trend. In other words, when the two are at extreme levels and begin to converge, look for a change in trend.

Note that in this discussion a change in trend does not necessarily mean that an upward trend will reverse to a downward trend. Sideways movement is also considered a "change."

Example

Figure 170 shows America Online, a 14-day Standard Error, and a 14-day r-Squared. The rapidly falling r-Squared, combined with the rising Standard Error, indicated that a change in the upward trend was occurring. This was borne out as America Online corrected over the next two months.

FIGURE 170



Calculation

The calculation for Standard Error is beyond the scope of this book. Standard Error can be calculated in a spreadsheet using the *steyx()* function. The A-to-Z Companion Spreadsheet (see page xvii) contains examples of both the brute force and the *steyx()* method.

STANDARD ERROR BANDS

Overview

Standard Error Bands, developed by Jon Anderson of Equis International, are a type of Envelope (see page 137) based on Standard Error (see page 311). They are similar in appearance to Bollinger Bands (see page 71), but they are calculated and interpreted quite differently. Where Bollinger Bands are plotted at Standard Deviation levels above and below a moving average, Standard Error Bands are plotted at Standard Error levels above and below the Linear Regression Indicator (see page 172).

The center line of the Standard Error Bands is simply the Linear Regression Indicator. The upper and lower bands are calculated by adding or subtracting the specified number of Standard Errors.

Interpretation

Because the spacing between Standard Error Bands is based on the Standard Error of the security, the bands widen when the volatility around the current trend (as defined by the Linear Regression Indicator) increases, and they contract when volatility around the current trend decreases.

Since Standard Error Bands are statistically based, other statistical indicators such as r-Squared, Standard Error, Linear Regression Indicator, etc., work well for trade confirmation.

Andersen notes the following characteristics of Standard Error Bands:

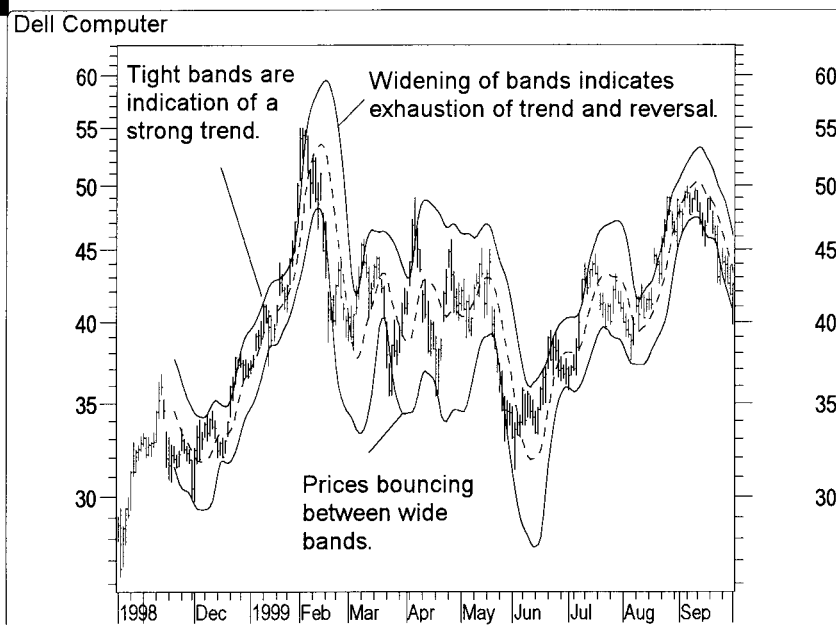
- Tight bands are an indication of a strong trend.
- Prices tend to bounce between the bands when the bands are wide.
- Tight bands followed by a widening of the bands may indicate the exhaustion of a trend and a possible reversal.

- When the bands reverse direction after an exhausted trend, prices tend to move in the direction of the bands.
- The r-Squared indicator works well in combination with Standard Error Bands. A high r-Squared value combined with tight bands confirms a strong trend. A low r-Squared value combined with wide bands confirms that prices are consolidating.

Example

Figure 171 of Dell Computer and 21-day Standard Error Standard Bands shows three of the five characteristics noted by Jon Anderson.

FIGURE 171



Calculation

The center line of the Standard Error Bands is a Linear Regression Indicator (see page 172). The upper band is calculated by adding the specified number of Standard Errors (see page 311) to the Linear Regression Indicator. The lower channel is calculated by subtracting the specified number of standard errors from the Linear Regression Indicator.

STANDARD ERROR CHANNEL

Overview

Standard Error Channels are calculated by plotting two parallel lines above and below an n -period Linear Regression Trendline (see page 176). The lines are plotted a specified number of Standard Errors (see page 311) away from the Linear Regression Trendline.

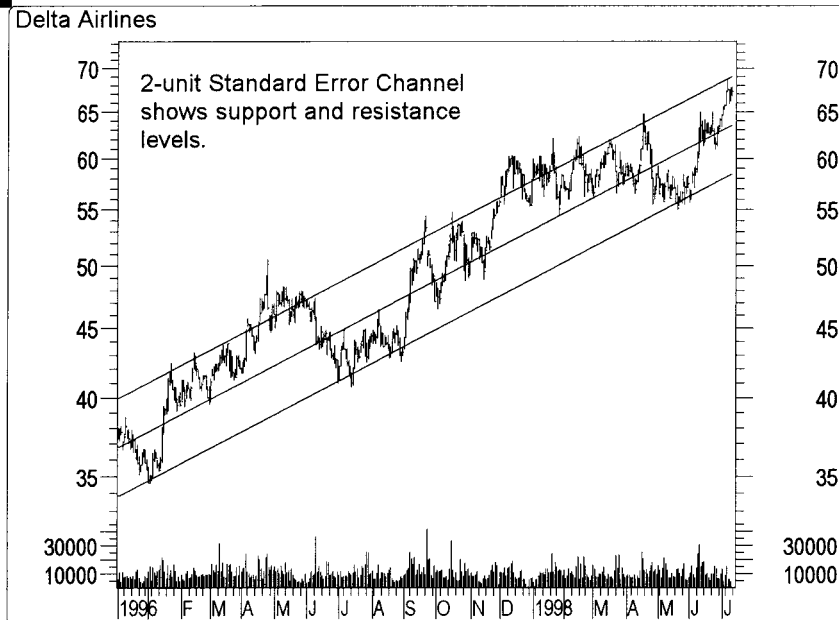
Interpretation

Standard Error Channels are similar to other channels (see Raff Regression Channels on page 284 and Standard Deviation Channels on page 310). The upper and lower bands should contain price movement. Extended moves outside of either band may indicate a change in trend.

Example

Figure 172 shows Delta Airlines and a 2-unit Standard Error Channel. The upper and lower channels acted as solid support and resistance for the 20 months displayed in the chart.

FIGURE 172



Calculation

The center line of the Standard Error Channel is a Linear Regression Trendline (see page 176). The upper channel is calculated by adding the requested number of Standard Errors to the Linear Regression Trendline. The lower channel is calculated by subtracting the requested number of Standard Errors from the Linear Regression Trendline.

STIX

Overview

STIX is a short-term trading oscillator that was published in *The Polymetric Report*. It compares the amount of volume flowing into advancing and declining stocks.

Interpretation

According to *The Polymetric Report* (see Table 73, also):

TABLE 73

Condition	Range
Extremely Overbought	greater than 58
Fairly Overbought	greater than 56
Fairly Oversold	less than 45
Extremely Oversold	less than 42

S

- STIX usually ranges between +42 and +58.
- If STIX gets as low as 45, the market is almost always a buy, except in a raging bear market.
- The market is fairly overbought if STIX rises to 56; and except in a new bull market, it's wise to sell if STIX goes over 58.
- Traders and investors should modify these rough rules to suit their own objectives.
- In normal markets, STIX rarely gets as high as 56 or as low as 45, so rigid use of these rules of thumb would keep you inactive most of the time. For active accounts, the rules might be made much less stringent.

Interpretation

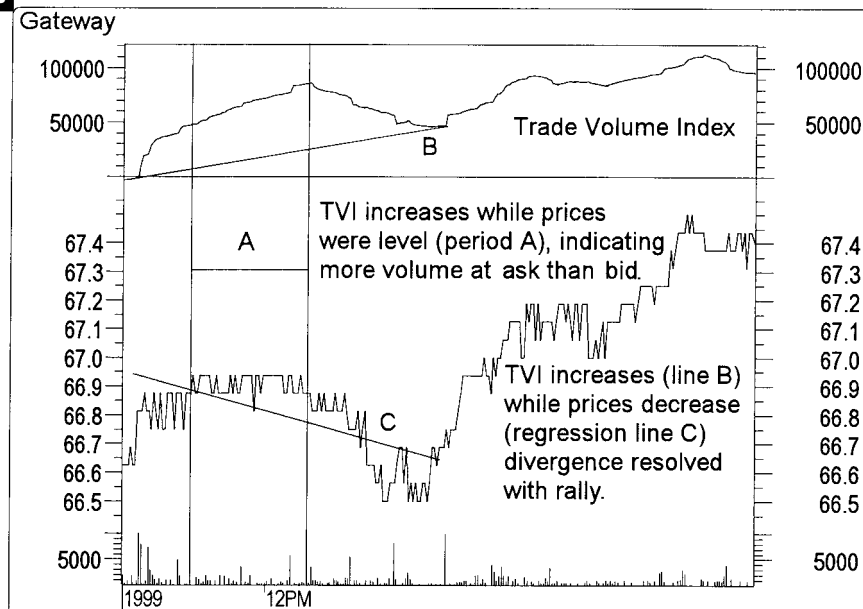
The TVI helps identify whether a security is being accumulated or distributed. When the TVI is trending up, it shows that trades are taking place at the asking price as buyers accumulate the security. When the TVI is trending down, it shows that trades are taking place at the bid price as sellers distribute the security.

When prices create a flat resistance level and the TVI is rising, look for prices to break out to the upside. When prices create a flat support level and the TVI is falling, look for prices to drop below the support level.

Example

Figure 185 shows Gateway's tick prices and the TVI. During the period labeled "A," prices remained at the same level while the TVI steadily increased. This indicated that more volume was occurring at the ask than at the bid. Also, there was a divergence between the TVI and Gateway's prices (Trendline B and Linear Regression Trendline C), which was resolved when the price moved up.

FIGURE 185



Calculation

The TVI is calculated by adding each trade's volume to a cumulative total when the price moves up by a specified amount and subtracting the trade's volume when the price moves down by a specified amount. The "specified" amount is called the Minimum Tick Value.

The calculations are very awkward in a spreadsheet, so I've skipped my customary table and explained the basic steps.

To calculate the TVI you must first determine if prices are being accumulated or distributed based on the Change and Minimum Tick Value:

$$MTV = \text{Minimum Tick Value}$$

$$\text{Change} = \text{Price minus the extreme price since direction changed}$$

If the Direction (defined below) is "Accumulate," the extreme price is the high since the direction changed to Accumulate. If the Direction is "Distribute," the extreme price is the low since the Direction changed to Distribute.

If Change is greater than MTV, then

Direction = Accumulate

If Change is less than $-MTV$, then

Direction = Distribute

If Change is less than or equal to MTV and

Change is greater than or equal to $-MTV$, then

Direction = Last Direction

Once you know the Direction, you can then calculate the TVI:

If Direction is Accumulate, then

$$TVI = \text{previous TVI} + \text{Volume}$$

If Direction is Distribute, then

$$TVI = \text{previous TVI} - \text{Volume}$$

TRENDLINES

Overview

One of the basic tenets put forth by Charles Dow in the Dow Theory (see page 123) is that security prices do trend. Trends are often measured and identified by “trendlines.” A trendline is a sloping line that is drawn between two or more prominent points on a chart.

- Rising trends are defined by a trendline that is drawn between two or more troughs (low points) to identify price support.
- Falling trends are defined by trendlines that are drawn between two or more peaks (high points) to identify price resistance.

Interpretation

A principle of technical analysis is that once a trend has been formed (two or more peaks/troughs have touched the trendline and reversed direction), it will remain intact until broken.

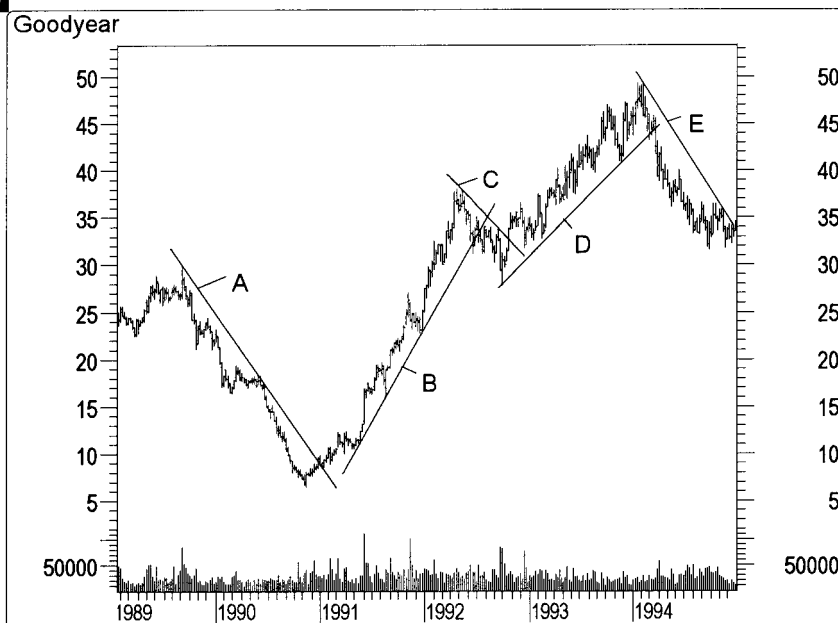
That sounds much more simplistic than it is! The goal is to analyze the current trend using trendlines and then either invest with the current trend until the trendline is broken or wait for the trendline to be broken and then invest with the new (opposing) trend.

The interpretation of trendlines is very similar to the interpretation of support and resistance concepts discussed earlier (see page 14).

One benefit of trendlines is that they help distinguish emotional decisions (“I think it’s time to sell”) from analytical decisions (“I will hold until the current rising trendline is broken”). Another benefit of trendlines is that they almost always keep you on the “right” side of the market. When using trendlines, it’s difficult to hold a security for very long when prices are falling just as it’s hard to be short when prices are rising—either way the trendline will be broken.

Example

Figure 186 shows Goodyear along with several trendlines. Trendlines A, C, and E are falling trendlines. Note how they were drawn between successive peaks. Trendlines B and D are rising trendlines. They were drawn between successive troughs in the price.

FIGURE 186

TRIX

Overview

TRIX is a momentum indicator that displays the percent rate-of-change of a triple exponentially smoothed moving average of a security's closing price. It is designed to keep you in trends equal to or shorter than the number of periods you specify.

Interpretation

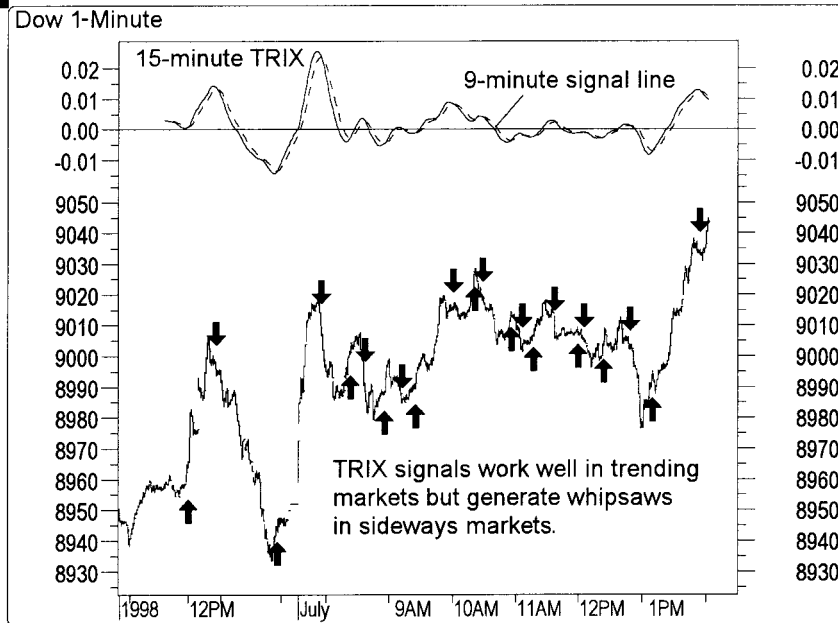
T

The TRIX indicator oscillates around a zero line. Its triple exponential smoothing is designed to filter out "insignificant" cycles (i.e., those that are shorter than the number of periods you specify).

Trades should be placed when the indicator changes direction (i.e., buy when it turns up and sell when it turns down). You may want to plot a 9-period moving average of the TRIX to create a "signal" line (similar to the MACD indicator, page 199), and then buy when the TRIX rises above its signal and sell when it falls below its signal.

Divergences between the security and the TRIX can also help identify turning points.

FIGURE 187



Example

Figure 187 shows a 1-minute chart of the Dow Industrials and a 15-minute TRIX (solid line) with a 9-minute “signal” moving average of the TRIX (dotted line). “Buy” arrows indicate when the TRIX rose above its signal line, and “sell” arrows indicate when it fell below its signal line. This method worked well when prices were trending, but it generated numerous false signals when prices were moving sideways.

Calculation

The TRIX is the 1-period percent rate-of-change (see page 267) of a triple exponentially smoothed moving average.

Table 79 illustrates the calculation of a 3-minute TRIX indicator:

- Column D is a 3-minute exponential moving average of the close. (See page 208 for details on calculating exponential moving averages.)
- Column E is a 3-minute exponential moving average of Column D.
- Column F is a 3-minute exponential moving average of Column E.
- To calculate Column G, first subtract the previous value in Column F from the current value in Column F. Next, divide this value by the current value in Column F. Finally, multiply this result by 100. This is the TRIX. Note that a 3-period TRIX isn’t valid until the seventh period.

TABLE 79

A	B	C	TRIX D	E	F	G
Date	Time	Close	3-period EMA of Close	3-period EMA of Col D	3-period EMA of Col E	1-period % change in Col F
06/30/98	7:35 AM	8,996.96				
06/30/98	7:36 AM	9,003.19				
06/30/98	7:37 AM	9,010.41	9,005.24			
06/30/98	7:38 AM	9,008.07	9,006.66			
06/30/98	7:39 AM	9,018.03	9,012.34	9,009.15		
06/30/98	7:40 AM	9,009.80	9,011.07	9,010.11		
06/30/98	7:41 AM	9,011.55	9,011.31	9,010.71	9,010.17	
06/30/98	7:42 AM	9,020.26	9,015.79	9,013.25	9,011.71	0.0171
06/30/98	7:43 AM	9,013.29	9,014.54	9,013.89	9,012.80	0.0121
06/30/98	7:44 AM	9,018.78	9,016.66	9,015.28	9,014.04	0.0137

TYPICAL PRICE

Overview

The Typical Price indicator is simply an average of each day's price. The Median Price (see page 190) and the Weighted Close (see page 362) are similar indicators.

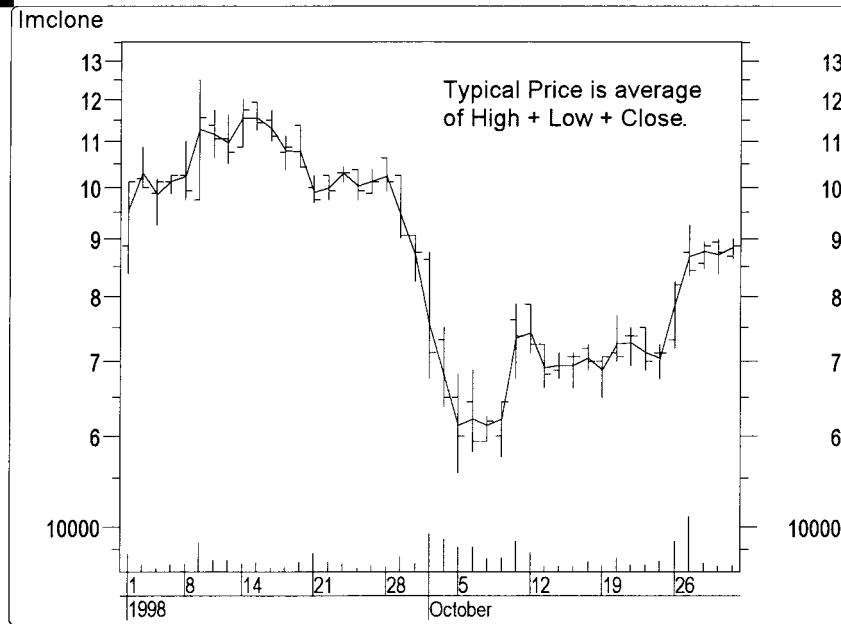
Interpretation

The Typical Price indicator provides a simple, single-line plot of the day's average price. Some investors use the Typical Price, rather than the closing price, when creating moving average penetration systems.

The Typical Price is a building block of the Money Flow Index (see page 197).

Example

Figure 188 shows the Typical Price indicator on top of Imclone's bar chart.

FIGURE 188

Calculation

The Typical Price indicator is calculated by adding the high, low, and closing prices together and then dividing by 3, as illustrated in Table 80. The result is the average (or typical) price.

$$\frac{(\text{High} + \text{Low} + \text{Close})}{3}$$

TABLE 80

TYPICAL PRICE					
A	B	C	D	E	F
Date	High	Low	Close	Column B plus Column C plus Column D	Column E divided by 3.0
09/01/98	10.1250	8.3750	10.1250	28.6250	9.5417
09/02/98	10.8750	10.0000	10.0000	30.8750	10.2917
09/03/98	10.1875	9.2500	10.1250	29.5625	9.8542
09/04/98	10.2500	9.8750	10.2500	30.3750	10.1250
09/08/98	11.0000	9.7500	9.9375	30.6875	10.2292
09/09/98	12.5000	9.7500	11.5625	33.8125	11.2708
09/10/98	11.7500	10.6250	11.0625	33.4375	11.1458
09/11/98	11.6250	10.5000	10.7500	32.8750	10.9583

ULTIMATE OSCILLATOR

Overview

Price oscillators typically compare a security's smoothed price with its price n periods ago. Larry Williams noted that the value of this type of oscillator can vary greatly depending on the number of time periods used during the calculation. So he developed the Ultimate Oscillator, which uses weighted sums of three oscillators, each of which uses a different time period. The three oscillators are based on Williams's definitions of buying and selling "pressure."

Interpretation

Williams recommended initiating a trade following a divergence (see page 36) and a breakout in the Ultimate Oscillator's trend. The following text summarizes these rules.

Buy when:

1. A bullish divergence occurs. This is when the security's price makes a lower low that is not confirmed by a lower low in the Ultimate Oscillator. (It is "bullish," because the indicator is forecasting higher prices.)
2. During the bullish divergence, the Ultimate Oscillator falls below 30.
3. The Ultimate Oscillator then rises above the highest point reached during the span of the bullish divergence. This is the point at which you buy.

Close long positions when:

1. The conditions are met to sell short (explained below), or
2. The Ultimate Oscillator rises above 50 and then falls below 45, or
3. The Ultimate Oscillator rises above 70. (I sometimes wait for it to then fall below 70.)

Sell short when:

1. A bearish divergence occurs. This is when the security's price makes a higher high that is not confirmed by a higher high in the Ultimate Oscillator.
2. During the bearish divergence, the Ultimate Oscillator rises above 50.
3. The Ultimate Oscillator then falls below the lowest point reached during the span of the bearish divergence. This is the point at which you sell short.